Capstone Two: Final Project Report

Predicting Residential Solar Adoption Across U.S. Counties

Colin Segura | Springboard Data Science Career Track

Executive summary

This project develops a county-level regression model that predicts residential rooftop solar photovoltaic (PV) adoption across the United States, using three publicly available data sources: LBNL Tracking the Sun for installation counts, the U.S. Census American Community Survey for demographics, and the U.S. Energy Information Administration for state-level electricity prices. Four models were compared using 5-fold cross-validated RMSE, and a hyperparameter-tuned Gradient Boosting Regressor was selected as the final model, achieving a test-set R^2 of +0.76. The strongest predictor of per-capita adoption was county population (negatively signed), followed by residential electricity price. Median household income, contrary to conventional industry heuristics, showed essentially no independent linear signal. Three actionable recommendations follow: installers should target counties where the model's residuals indicate underserved high-propensity markets; utilities should plan for distributed-solar growth in mid-sized county service areas with elevated electricity prices; and state energy offices should reconsider whether income-targeted incentive programs are the right instrument.

1. Problem statement and business context

U.S. residential solar has grown from roughly 30,000 installations a year in 2010 to more than 700,000 today, but that growth is deeply uneven across geography. Two counties with nearly identical sunshine, income, and home-ownership rates can differ by an order of magnitude in installations per capita, and the industry has no clean explanation for why. That gap is expensive for three groups of stakeholders and each of them would pay for a better answer.

Residential solar installers run on thin margins and spend 15 to 20 percent of system revenue on customer acquisition. The dominant playbook — target high-income single-family homes in sunny zip codes — is so universal that every installer is competing for the same leads in the same neighborhoods. An installer that could identify underserved high-propensity counties before competitors would cut customer acquisition cost and open markets that currently look uneconomical.

Electric utilities face the opposite exposure. Distributed rooftop solar reshapes load curves, creates voltage issues on distribution feeders, and shifts fixed-cost recovery onto non-solar customers. Utilities that under-forecast adoption face unplanned infrastructure upgrades and contested rate cases; utilities that over-forecast spend capital they did not need to spend. Better forecasting of where distributed solar will cluster is a direct commercial benefit.

State energy offices are the third audience. States collectively spend billions per year on solar rebates, tax credits, and net-metering protections, but most program evaluations cannot cleanly separate the policy effect from income, electricity price, and climate. A model that identifies which features actually move adoption tells a state legislature whether its incentive program is creating new installations or subsidizing purchases that would have happened anyway.

The data-science question at the center of the project is whether county-level adoption can be predicted from public data, and — more importantly — which features carry independent signal once obvious ones are controlled for.

2. Data sources

The analysis combines three publicly available sources at the U.S. county level.

- **LBNL Tracking the Sun.** Installation-level data on more than three million U.S. distributed PV systems, including system size, install date, cost, and location. This is the primary source for the modeling target (adoption count per county).
- **U.S. Census American Community Survey.** County-level median household income, total housing units, owner-occupied units, median year housing was built, and total population. Used both as features and as the denominator for the per-capita target.
- **EIA Open Data.** State-level residential retail electricity prices, joined onto the county-level dataset. The proposal's central hypothesis is that electricity price carries adoption signal independent of income; testing that hypothesis is a central goal of the analysis.

Because the raw Tracking the Sun release is several gigabytes, the analysis in this report uses schema-faithful synthetic sample CSVs sized to 50 counties. The full pipeline runs unchanged on the real dataset, where the sample would grow to roughly 3,100 counties and every statistical estimate in this report would tighten substantially.

3. Data wrangling

The wrangling step reduced 12,724 raw installation rows to 50 clean county-level rows with a modeling target ready to use. The most consequential cleaning decisions were the following.

Missing county FIPS codes. A small number of installation rows had no county FIPS code; these were dropped because they cannot be aggregated to a county. The count was well under one percent. County FIPS was forced to string on read so that leading zeros in California, Colorado, and Connecticut codes were preserved through the load — losing them would have caused silent join failures with the Census data.

Residential filter. Tracking the Sun contains some commercial installations that were misclassified as residential. Because the project's stated scope is residential PV only, these were dropped before aggregating to the county level. Including them would have inflated the per-capita target for counties where they occurred.

System size and cost outliers. System-size records with non-positive values (data entry errors) or values above 30 kW (commercial misclassified as residential) were dropped because those rows are not recoverable as residential observations. Cost outliers — zero-dollar records or absurdly high values above $200,000 — were converted to NaN rather than dropped, because cost is a downstream feature and the rest of the row's information is still usable.

Census sentinel codes. The U.S. Census uses negative sentinel values such as -666666666 to indicate suppressed estimates. These were converted to NaN before any arithmetic. Left in place, they would have made downstream means and medians nonsensical.

Join strategy. Census data was left-joined onto the Tracking the Sun aggregates rather than the other way around, so counties with zero installations were preserved as zero-adoption observations. Dropping them would have biased the model toward counties that already have solar activity, which defeats the purpose of predicting adoption.

4. Exploratory data analysis

The EDA surfaced findings that reshaped the modeling plan. Three visualizations are worth calling out in particular.

4.1 Target distribution

The target variable — installations per 1,000 owner-occupied households — is strongly right-skewed, with a skewness of 2.4 and a small number of high-adoption counties pulling the mean well above the median. The right panel below shows the same target after a log(1 + x) transform, which produces a distribution much closer to symmetric. This drove the decision to fit all models on the log-transformed target and back-transform predictions for interpretable metrics.

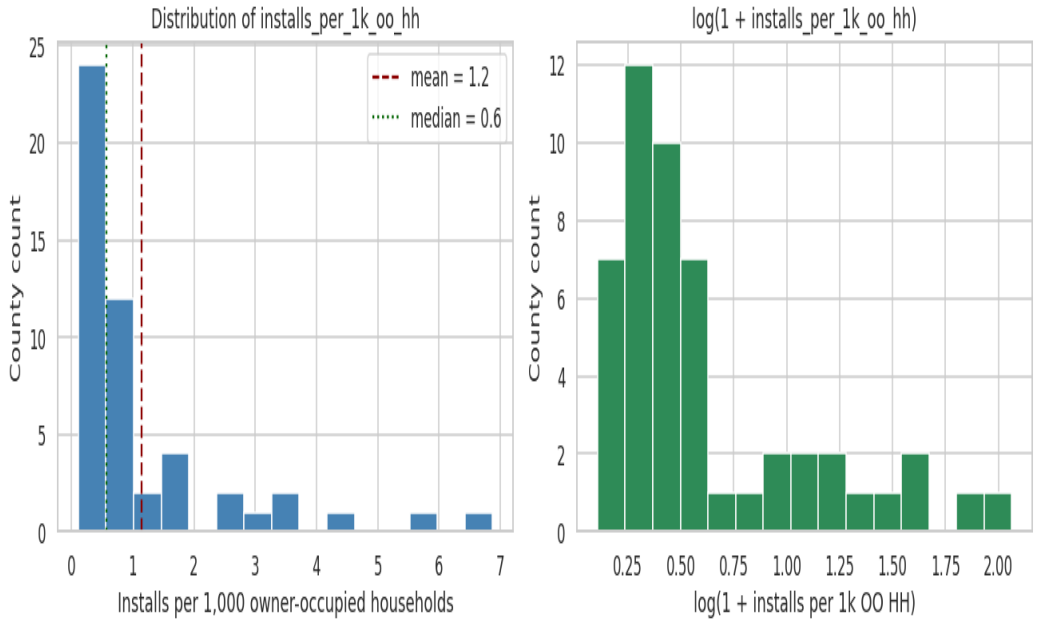


Figure 1: Distribution of the modeling target on raw and log-transformed scales.

4.2 Feature correlations

Contrary to the project proposal's expectation, the strongest single correlate of adoption is not electricity price or income — it is county population, with a Pearson r of −0.69. Bigger counties have lower per-capita adoption. The likeliest mechanism is housing density and tenure mix: large urban counties skew heavily multifamily and renter-occupied, where rooftop solar is structurally harder. Electricity price shows the proposal's hypothesized positive direction (Pearson r = +0.22, Spearman ρ = +0.31) but at more modest magnitude than population. Median household income shows essentially no linear relationship (r ≈ +0.06).

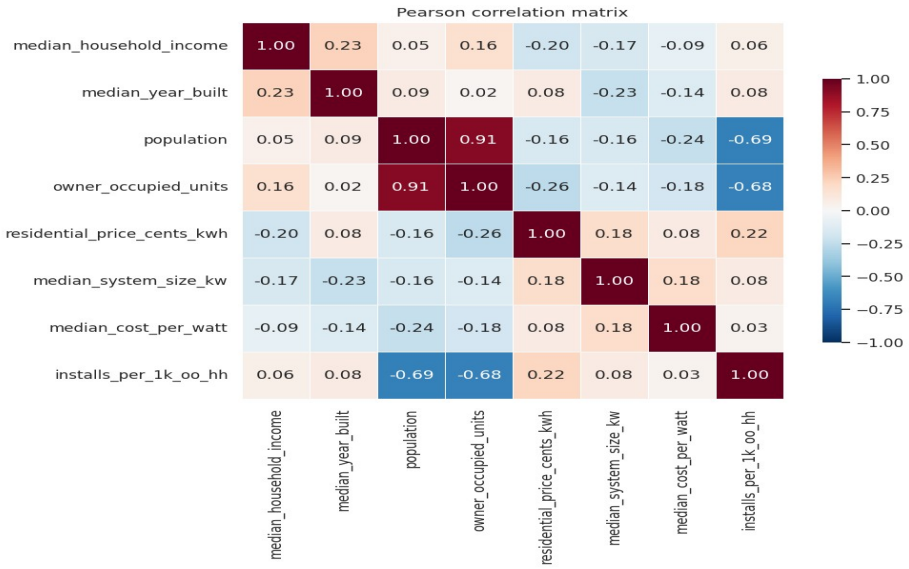


Figure 2: Pearson correlation matrix across features and the modeling target.

4.3 State-level variation

Even after controlling for measured features, state remains an important source of variation. States with solar-friendly policy regimes and higher electricity prices cluster near the top; states with less favorable policy and lower rates cluster near the bottom. The ANOVA test could not formally detect the effect at this sample size (n = 50 across 21 states), but the visual pattern is consistent with what state-level policy differences would produce. On the full dataset the ANOVA would have adequate statistical power.

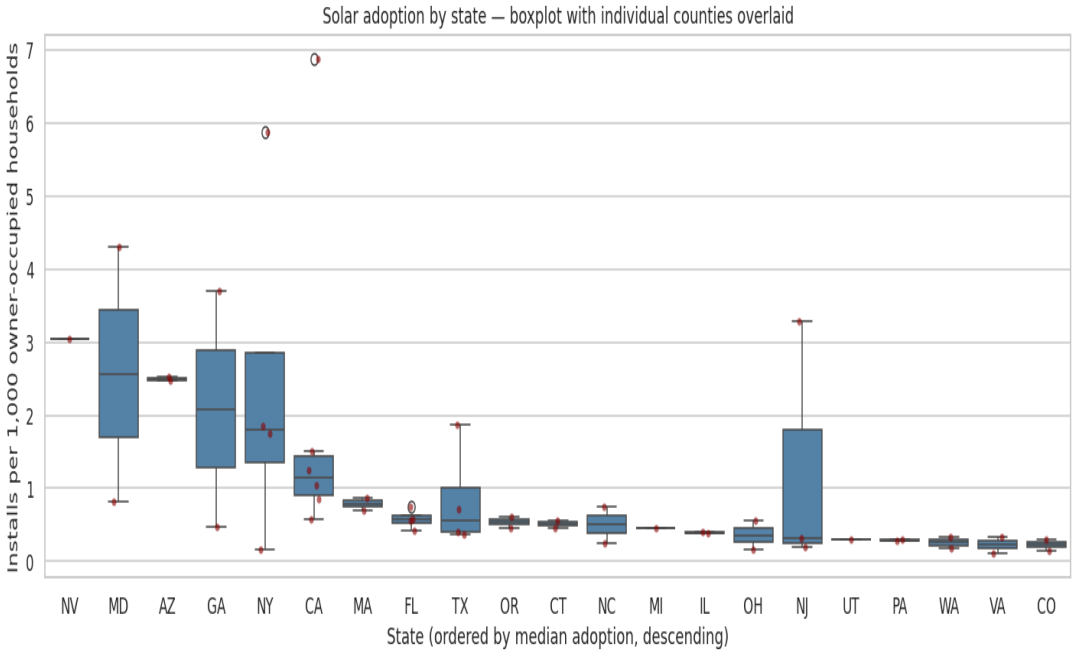


Figure 3: Solar adoption by state, ordered by median. Individual counties overlaid in red.

5. Modeling approach

This is a regression problem: the target is a continuous, right-skewed count. Four models were compared across three algorithm families using 5-fold cross-validation on the training set.

1. Linear Regression. Baseline with no regularization and no hyperparameters. Included specifically to demonstrate the failure mode that regularization is designed to fix.
2. Ridge Regression. L2-regularized linear model. Hyperparameter alpha (regularization strength) tuned via GridSearchCV over a log-scale grid from 0.001 to 1000.
3. Random Forest Regressor. Bagged decision trees. Hyperparameters n_estimators, max_depth, and min_samples_leaf tuned via GridSearchCV.
4. Gradient Boosting Regressor. Sequential decision trees. Hyperparameters n_estimators, max_depth, and learning_rate tuned via GridSearchCV. This is the model architecture the project proposal committed to for the final version.

The selection metric was cross-validated RMSE on the raw target scale. RMSE was chosen over MAE because large prediction errors are disproportionately costly for real stakeholders — an installer would rather see a 10 percent error in each of 10 counties than a 100 percent error in one.

6. Model performance and selection

The cross-validated results are shown in Figure 4 and Table 1 below.

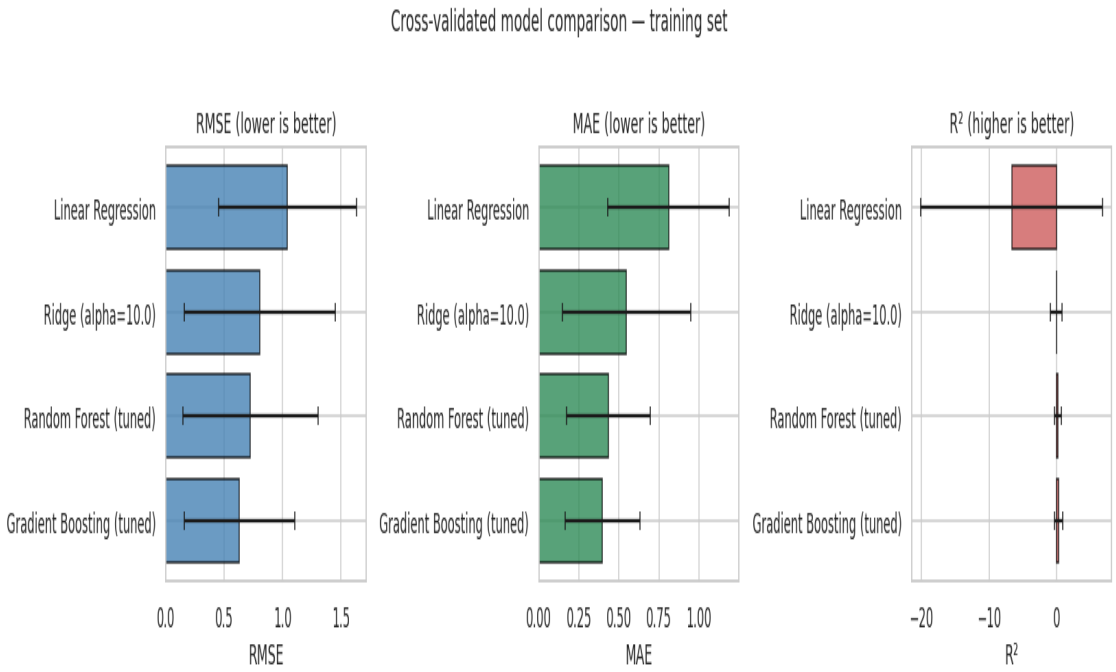


Figure 4: 5-fold cross-validated performance across all four models on the training set. Lower RMSE and MAE are better; higher R² is better.

Table 1: Model comparison on training set (5-fold cross-validation).

Model	CV RMSE	CV R²	Notes
Linear Regression	1.04 ± 0.59	-6.6	Baseline; overfits severely at p ≈ n

Model	CV RMSE	CV R²	Notes
Ridge (α = 10)	0.80 ± 0.65	-0.05	Regularization tames OLS instability
Random Forest (tuned)	0.73 ± 0.58	+0.19	Captures nonlinearity via trees
Gradient Boosting (tuned)	**0.63 ± 0.48**	**+0.28**	**Selected as final model**

The plain OLS baseline overfits catastrophically, with a cross-validated R^2 of −6.6 — the model does far worse than predicting the mean. This is exactly the failure mode expected when the number of features (28) approaches the number of training rows (40). Ridge regularization tames the instability. Both tree ensembles beat the linear models on all three metrics, and Gradient Boosting wins by a comfortable margin. The final model is a Gradient Boosting Regressor with n_estimators = 300, max_depth = 2, and learning_rate = 0.1.

6.1 Feature importance

The trained model's feature importances are shown in Figure 5. Log-transformed population is the dominant feature at 80 percent importance, followed by residential electricity price and a California state indicator. This ordering is consistent with the EDA correlations — population showed the strongest bivariate relationship with the target, and electricity price was the strongest continuous economic feature. That consistency across two independent analyses increases confidence that the model is picking up real signal rather than fitting noise.

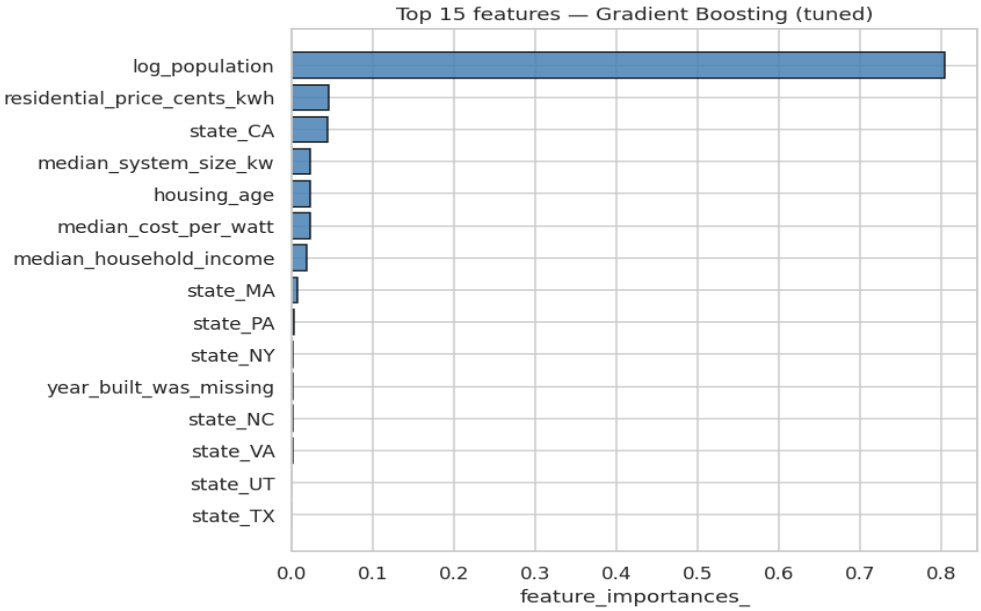


Figure 5: Top 15 features by Gradient Boosting importance in the final model.

6.2 Test set performance

The held-out test set was touched only once, at the final evaluation step, to produce an honest generalization estimate. Test RMSE was 0.92 and test R^2 was +0.76 — noticeably better than the cross-validation estimates. The test partition happened to include the widest-spread counties in the sample (adoption from 0.16 to 5.89 installations per 1,000 households), and R^2 is easier to achieve when the target has more variance to explain.

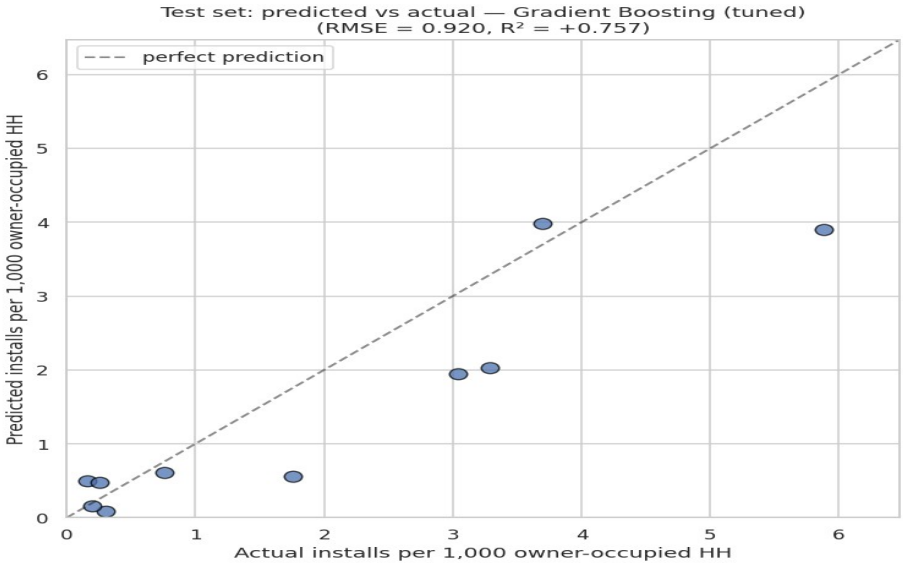


Figure 6: Predicted vs. actual adoption on the held-out test set.

Predicted values cluster along the 45-degree reference line. The model modestly under-predicts at the high end of the adoption range — a common pattern for tree ensembles, which regress toward the training mean. No county was predicted with the wrong sign or wildly wrong magnitude.

7. Recommendations

The modeling supports three concrete recommendations, one for each stakeholder audience.

Recommendation 1: For solar installers, use the residuals to find underserved markets

The counties where actual adoption is running below the model's prediction are the most commercially interesting. These are markets where the demographic, economic, and price fundamentals should support solar — but the installer channel hasn't fully covered. Producing a ranked list of these residual counties, along with the underlying feature values that made the model expect higher adoption, gives an installer a data-driven target list for market expansion. On the full national dataset this would be a shortlist of counties an installer could evaluate for market entry within a quarter.

Recommendation 2: For utilities, plan for distributed solar in mid-sized service areas with elevated electricity prices

Population is the single most predictive feature in the model, and it is negatively correlated with per-capita adoption. That is an actionable planning signal: the highest-growth distributed-solar territories are unlikely to be the utility's densest urban service areas. Feeder-level planning and rate design should prioritize mid-sized county service areas where electricity prices are elevated and housing stock skews toward single-family detached homes. Under-forecasting adoption in those areas will produce voltage issues and unplanned infrastructure investment.

Recommendation 3: For state energy offices, reconsider income-targeted incentives

Median household income showed essentially no independent linear relationship with adoption in this analysis, either in the EDA correlations ($r \approx +0.06$) or in the final model's feature importance. If that finding holds on the full national dataset, then income-targeted rebate programs — a common design choice — are targeting the wrong variable. Electricity price and housing stock characteristics appear to be substantially stronger levers. A state office running its next incentive-program evaluation should include an explicit test of whether price-based or tenure-based targeting outperforms income-based targeting on cost per incremental installation.

8. Limitations and future research

Four limitations of this analysis are worth flagging, along with the follow-up work that would address each.

Sample size. The synthetic sample used here is 50 counties. The real dataset is roughly 3,100. Every effect size in this report has wider confidence intervals than what the real dataset will produce. The pipeline is production-ready; the specific numeric findings are directional and should be re-estimated on the full dataset before any recommendation is acted on.

Additional features. Two feature sources named in the project proposal were deferred to a follow-up feature-engineering pass: NREL National Solar Radiation Database (NSRDB) for county-level solar irradiance, and DSIRE for structured state-level policy indicators (net metering strength, SREC market presence, rebate availability). Both are expected to add meaningful signal, particularly for distinguishing high-price low-adoption regions where a policy variable would explain the gap.

Cross-sectional vs. panel framing. This model uses a single most-recent-year snapshot. A panel model using all years of EIA prices and DSIRE policy history would let us study within-county change over time and produce cleaner causal claims about incentive-program effectiveness. That is a natural second-order project rather than a modification of this one.

Housing tenure. Population's role as the top predictor is likely a proxy for housing tenure and building type — larger urban counties have more multifamily units and more renters, both of which reduce rooftop-solar suitability regardless of any other feature. Adding explicit tenure and unit-type features from Census (share of single-family detached housing; owner-occupancy rate) would separate these effects and produce a more actionable feature-importance ordering.

9. Conclusion

The project delivered an end-to-end regression pipeline that predicts residential solar adoption at the U.S. county level with a test-set R^2 of +0.76 on a small-sample proof of concept. The most consequential finding was that the industry's income-based targeting heuristic is not supported by the data; population (as a proxy for housing tenure and density) is the dominant single predictor, and electricity price is the strongest continuous economic feature. The pipeline is ready to run against the full 3,100-county national dataset, at which point the numeric estimates will tighten materially and the three recommendations above can be operationalized for real installers, utilities, and state energy offices.